Course Project: Final Report

DSE6211

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Executive Summary

Introduction

To assist ABC Hotels identify reservations with a high risk of cancellation, several machine learning models were developed and evaluated using a dataset of more than 36,000 records of past reservation data, provided by the company. This report will include an outline of the project approach, data and feature processing steps, a review of the comparison between three models, and identification of selected model.

Objective

The primary objective of this analysis is to provide a risk framework against which the company can compare new reservation data. The framework will allow ABC Hotels to identify high risk reservations so that cancellation mitigation techniques can be used to reduce the cancellation risk.

Methodology

Three feedforward neural network models were developed to perform supervised learning against the provided data set. These models were analyzed and compared to one another for accuracy, negative and positive result rates, calibration, and speed of processing. One model was selected and will be presented for recommendations and next steps.

Key Findings

All three feedforward neural networks delivered remarkably similar levels of accuracy, result rates, calibration and test results. The main difference between the three models was the processing time. Therefore, the model with the shortest processing time was selected as the most desirable model for this project.

Result

ABC Hotels will be able to use Model 1 as the model for identifying high risk customers and applying mitigation techniques.

Conclusion

Based on the past customer reservation data provided by the company, Model 1 was identified as the model with the most desirable outcome and usefulness.

Approach and Data

Approach

All developed supervised learning models resulted in cancellation data risk values that fell between 0 and 1. These values can be interpreted as the probability of cancellation, with zero having no risk and one having the most-probable risk of cancellation. In this case, the target feature will be “booking_status”, and the classification models predict the probability of ”booking_status” being “not_canceled”.

Questions and Goal

Questions to be answered as part of this project include the following:

- Can we determine the cancellation probability of a booking based on the available data?
- What other data would be helpful in improving the risk calculation?
- What are some suggested risk mitigation techniques to apply to high-risk bookings?

The primary goal of this project is to reduce the number of anticipated booking cancellations. As a secondary goal, the selected model will limit the number of false positives to save ABC Hotels from unnecessarily spending resources on bookings that are not high-risk.

Data

ABC Hotels provided a dataset of 36,238 records of previous reservation data. The dataset was determined to be complete, not missing values, and not having unexpected values. There were no records with null values. All but one of the original 17 features will be used as part of the model development. Below is a table for the training dataset that identifies each feature, whether used or eliminated, and its required data encoding steps.

Column	Data Processing and Encoding Steps
Booking_ID	Eliminated; this is a customer identifier that provides value to the risk calculation
no_of_adults	
no_of_children	
no_of_weekend_nights	
no_of_week_nights	
type_of_meal_plan	OneHotEncoder converted into four features; original feature eliminated
required_car_parking_space	
room_type_reserved	OneHotEncoder converted into seven features; original feature eliminated
lead_time	
arrival_date	Eliminated after converting first to "season" and then into four "season" features via OneHotEncoder
market_segment_type	OneHotEncoder converted into five features; original feature eliminated
repeated_guest	
no_of_previous_cancellations	
no_of_previous_bookings_not_canceled	
avg_price_per_room	
no_of_special_requests	
season	Four features created from "arrival_date" values
booking_status	This is the target label; performed label encoding to turn the categorical values into numeric values. 0 = does not cancel 1 = will cancel

Figure 1: Dataset Description, Processing, and Encoding

Feature Engineering

"Booking_ID" will not be used in model development as it is an identifier and has no predictive value. Another feature, "arrival_date" will be recast into the appropriate "season" and subsequently eliminated from the dataset. The following features will be converted into additional features using the OneHotEncoder() from sklearn.processing:

- type_of_meal_plan
- room_type_reserved
- market_segment_type
- season

The dataset of 36,238 records was split into a training set of 27,178 records and a test set of 9,060 records in a 75:25 ratio. The training set was further split into a training-only dataset and a validation dataset in an 80:20 ratio. The order of the features was modified to put all numeric features in consecutive columns and move non-numeric features to the end columns. This was done to simplify the column-binding steps further down in the model development process.

The target feature of “booking_status” was encoded using LabelEncoder() from sklearn.processing. All features were centered and scaled using StandardScaler(), also from sklearn.processing.

An additional feature engineering step was to incorporate Principal Component Analysis on 9 numeric features to reduce the number of features analyzed. The 9 features included:

- no_of_adults
- no_of_children
- no_of_weekend_nights
- no_of_weekday nights
- lead_time
- no_of_previous_cancellations
- no_of_previous_bookings_not_cancelled
- avg_price_per_room
- no_of_special_requests

Machine Learning Methods

The project entailed creating three feedforward neural network models using Python with Pytorch, pandas, sklearn, seaborn, and matplotlib libraries. All three models are Sequential models with varying numbers of hidden layers and nodes. The activation function for all hidden layers in all models is the ReLU activation function that introduces non-linearity to the model. The function used in the output layer in all models is the Sigmoid function.

The optimizer used in all models is the Adam optimizer. Note that other optimizers were attempted but were not helpful and were quickly changed to Adam. There are varying numbers of epochs in each model. The loss function used by all models is the BCELoss function since these are binary classification models.

Returning to the discussion about PCA, the principal component analysis revealed that 80% of the variance could be explained with the first 6 components, so the original observations were projected onto those components. The projection was repeated with the test set and tensors were created from the outcomes. The scree plot below breaks down the amount of explained variance by component, and the cumulative explained variance plot shows that the first 6 components cumulatively explain 80% of the variance. By eliminating 3 components, the models were able to run more efficiently with less complexity.

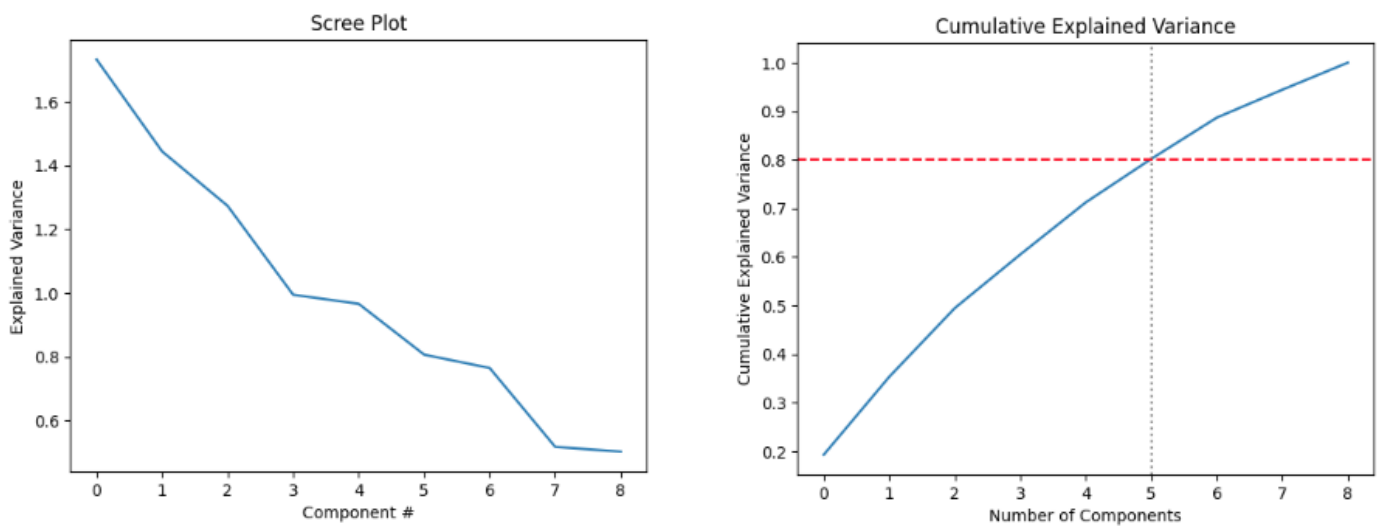

Figure 2: Explained Variance and Cumulative Explained Variance

Detailed Findings and Evaluation

Model Descriptions

Model 1

The first neural network model developed had two hidden layers, one with 20 nodes and the other with 35 nodes. A learning rate of 1e-3 was used with the Adam optimizer. There were 400 epochs used.

Model 2

The second neural network model developed also had two hidden layers, one with 75 nodes and the other with 20 nodes. A learning rate of 1e-2 was used with the Adam optimizer. There were 100 epochs used.

Model 3

The final neural network model developed had three hidden layers with 15, 40, and 25 nodes, respectively. A learning rate of 1e-4 was used with the Adam optimizer. There were 2000 epochs used.

Model Evaluations

A number of methods were used to evaluate the performance of the three models. It should be noted that the methods were considered as a group, rather than any one method being preferred over another one.

Accuracy

For purposes of machine learning, accuracy is described as a measurement of how often a model correctly predicts the outcome (Evidently AI Team, 2025). For each model, accuracy was calculated using the accuracy_score function in the sklearn.metrics library.

Model 1	Model 2	Model 3
83.36%	80.77%	82.22%

Figure 3: Model Accuracy Comparison

Each model had a similar accuracy rate in the low- to mid- 80% range, with Model 1 having the best accuracy.

Confusion Matrix

The confusion matrix for machine learning is used for classification models and compares the model's results against the actual results (Geeks For Geeks, 2025).

Model 1

Predicted	0	1
Actual		
0	1991	938
1	570	5561

Model 2

Predicted	0	1
Actual		
0	1903	1026
1	716	5415

Model 3

Predicted	0	1
Actual		
0	1974	955
1	656	5475

Figure 4: Model Confusion Matrix Comparison

Each model had similar results in the confusion matrix, so the next step was to compare the false positive rates of each model. Minimizing the false positive rate will help ABC Hotels minimize wasted resources on lower risk bookings that were predicted to be high risk.

Model 1	Model 2	Model 3
14.43%	15.93%	14.85%

Figure 5: Model False Positive Rate Comparison

Again, all models were very similar, and Model 1 has the lowest false positive rate.

ROC Plots

The ROC curve is a visual representation of model performance across thresholds (Google Developers, 2025). Below are the ROC curves for each model.

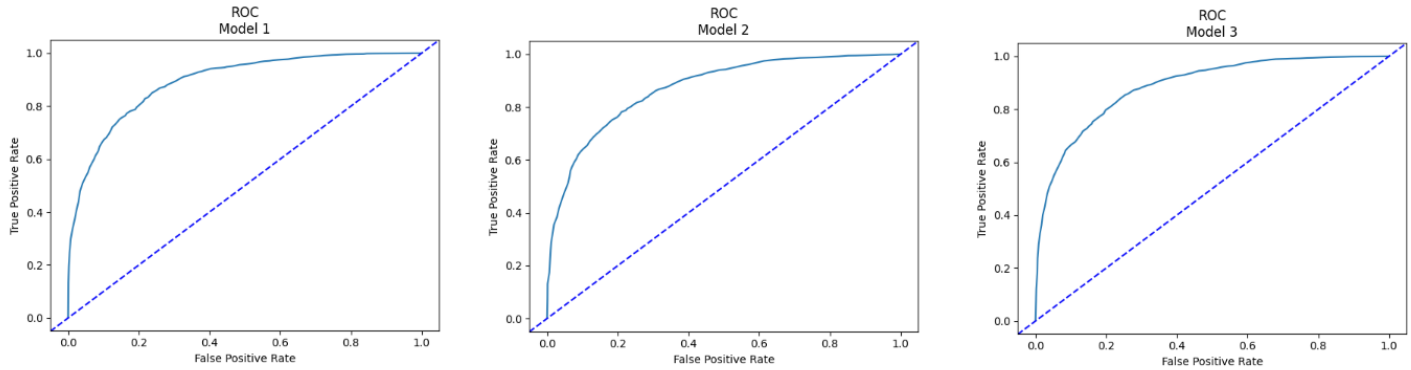

Figure 6: Model ROCAUC Comparison

The models' ROC curves and AUC were very similar to one another.

Jitter Plots

These jitter plots were used to visually represent how many positive vs. negative predicted results were identified by the models.

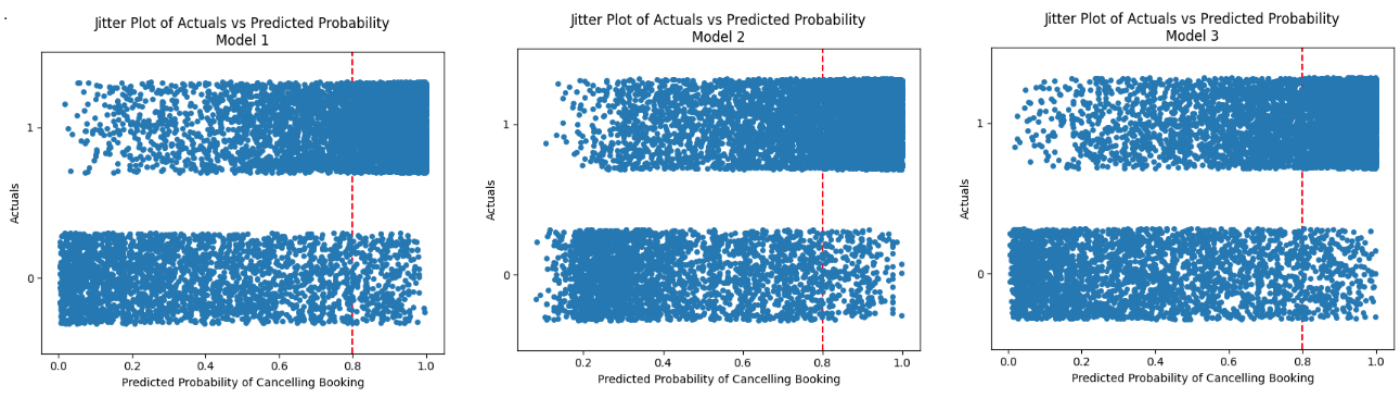

Figure 7: Model Jitter Plot Comparison

As with previous evaluation methods, the models evaluated to be very similar using the jitter plots.

Calibration Curves

Calibration curves identify discrepancies between the predictions and the actual data probabilities (Wijaya, 2023). Here is a comparison of the calibration curves of these three models.

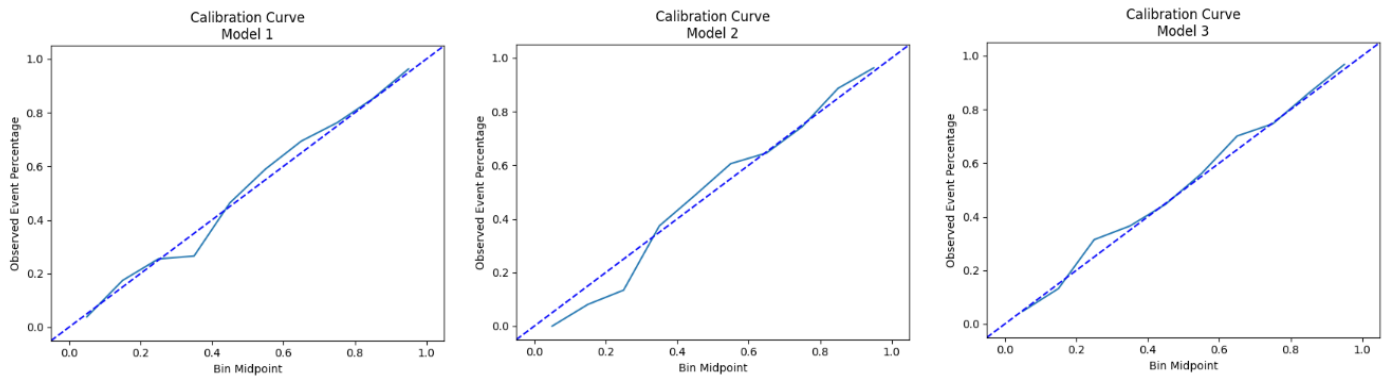

Figure 8: Model Calibration Curve Comparison

The models are again very similar, with Model 3 having a slightly better calibration by a small margin.

Processing Time

While processing time is not usually considered an evaluation method, it can come into play as measure of efficiency when two or more models are being considered for a project and are otherwise evaluating very similarly, as is the case here. So, for purposes of completeness, below is the amount of time each model took to complete. Note that there were no differences in processing environments for each model.

Model 1	Model 2	Model 3
9.11 seconds	12.76 seconds	49.21 seconds

Figure 9: Model Processing Time Comparison

In this case, Model 1 appears to be the most efficient model, with all other factors being equal.

Analytical Findings

Model Selected

Model 1 has been selected as the best model of the three. All models produced very similar results and performed very similarly. Model 1 has slightly better accuracy and a slightly lower False Positive rate, but neither difference is striking enough to make the model stand out from the others. The major difference between Model 1 and the other models is the short amount of time required to process the model. Model 2 was only three seconds slower and could arguably still be used as the selected model. Model 3, however, was much slower and required 49 seconds to process the same data set.

Recommendations

Business Questions Answered

This project aimed to answer the following questions:

Can we determine the cancellation probability of a booking based on the available data?

Yes, several reasonably good feedforward neural network models were developed that can predict the probability of a cancelled hotel reservation. The selected model can be used as a guide to identify the cancellation risk of future bookings.

What other data would be helpful in improving the risk calculation?

Two other datapoints could potentially improve a model's capability to predict cancellation risk. The first suggested improvement is to capture the source of the booking. While there is a datapoint called "market_segment_type" in the provided data, a slightly different datapoint of origins may provide a different perspective. Customers have several ways to reserve a hotel room including directly calling the hotel, booking through the hotel's website, using a travel agent to

book a room, and booking the room through a third party (Expedia, Hotels.com, etc.). Knowing the origin of reservation could potentially identify whether a reservation method is too easy to book and cancel with, as may be the case with third party sites

The second suggested data collection improvement is to capture the reason for the cancellation. To simplify the data collection, a set of common reasons could be used. Suggested reasons might include illness, event cancellation, changes in schedule, finding a better room rate elsewhere, finding a hotel with better services or amenities, fraudulent reservations or other reasons. While some of these cancellation causes are out of ABC Hotel's control, it would be interesting to see if there is a trend regarding room rates or hotel services and amenities. This data may suggest that a review of the hotel's competitiveness is warranted.

What are some suggested risk mitigation techniques to apply to high-risk bookings?

The first suggested risk mitigation technique is for ABC Hotels to directly contact the high-risk customer. This technique might include calling or emailing the customer and engaging them in a conversation during which the hotel re-emphasizes the hotel's services and amenities, offers a unique unadvertised service to this customer, or offers a discount on an additional night's stay.

The second technique is a bit broader in nature. Building customer loyalty takes resources and time but can provide a nice return on investment. Steps to building customer loyalty could include actively getting to know the customer through personalized surveys and communications, offering discounts to repeat customers, and creating a loyalty program with private unique incentives.

Goal Achievement

Two goals were identified for this project. The primary goal was to provide a framework that helps ABC Hotels reduce the number of anticipated reservation cancellations. The selected model can provide a reasonably accurate method for identifying high-risk cancellations, thereby affording ABC Hotels the opportunity to apply risk mitigation techniques to reduce the cancellation risk.

A secondary goal was to identify a model with the lowest false positive rate to help the company efficiently reduce high risk cancellations without wasting time and money on customers who were not going to cancel their reservations in the first place. The selected model has the lowest false positive rate of the three options.

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Appendix 1

Please refer to FinalProject_Take1.pdf to view the notebook for Model 1

Appendix 2

Please refer to FinalProject_Take2.pdf to view the notebook for Model 2

Appendix 3

Please refer to FinalProject_Take3.pdf to view the notebook for Model 3